Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 35: Approved Solutions

Combinations

Covers Chartrand & Zhang, Section 8.4

Status of this set

Ungraded practice. Solutions released with it. Show the expression.

Part I: How to Practice This Material

The two formulas, and the one question

$$P(n,r) = \frac{n!}{(n-r)!} \qquad C(n,r) = \binom{n}{r} = \frac{n!}{r!\,(n-r)!} = \frac{P(n,r)}{r!}$$

Ask: does reordering change the outcome? Yes ⇒ permutation. No ⇒ combination.

Two shortcuts

$\binom{n}{r} = \binom{n}{n-r}$, so compute whichever is easier. And when a condition is awkward — "at least one" — count the complement instead.

Three mistakes to expect

1. **Using P where C belongs**, over-counting by $r!$.
2. **Adding when the stages are independent**, or multiplying when the cases are exclusive.
3. **Computing "at least one" by adding overlapping cases.** Use the complement.

Part II: Practice Problems

Section A — Which is it?

1. For each, state whether it is a permutation or a combination, and compute it.

(a) Choosing 3 pizza toppings from 8.

(b) Choosing a batting order for 9 of 12 players.

(c) Choosing 5 books from 11 to take on a trip.

Approved solution

(a) Combination. Toppings are unordered: $\binom{8}{3} = 56$.
(b) Permutation. A batting order is a sequence: $P(12, 9) = \frac{12!}{3!} = 79{,}833{,}600$.
(c) Combination. A set of books: $\binom{11}{5} = 462$.

The test is always the same. Swapping two toppings gives the same pizza; swapping two batters gives a different order.

Section B — Combining principles

2. A committee of 4 is chosen from 6 seniors and 5 juniors.

(a) How many committees are possible?

(b) How many have exactly 2 seniors?

(c) How many have at least 1 junior?

Approved solution

(a) $\binom{11}{4} = 330$.
(b) Choose 2 seniors *and* 2 juniors — Multiplication Principle: $\binom{6}{2}\binom{5}{2} = 15 \cdot 10 = 150$.
(c) By complement: all committees minus those with no junior (all seniors): $330 - \binom{6}{4} = 330 - 15 = 315$.

Part (c) the direct way would be 1, 2, 3, or 4 juniors — four cases to add. The complement is one subtraction. **"At least one" almost always means complement.**

3. From a standard 52-card deck, how many 5-card hands

(a) consist entirely of hearts?

(b) contain exactly 3 hearts?

(c) contain at least one ace?

Approved solution

(a) $\binom{13}{5} = 1287$.
(b) Three hearts *and* two non-hearts: $\binom{13}{3}\binom{39}{2} = 286 \cdot 741 = 211{,}926$.
(c) Complement: $\binom{52}{5} - \binom{48}{5} = 2{,}598{,}960 - 1{,}712{,}304 = 886{,}656$.

In (b), after choosing the hearts the remaining cards must come from the 39 non-hearts — otherwise you would be counting hands with four hearts as if they had three.

Section C — Recognising a hidden choice

4. Count each.

(a) Bit strings of length 12 with exactly five 1s.

(b) Paths on a grid from $(0,0)$ to $(5,3)$ moving only right or up.

(c) Handshakes if everyone in a group of 8 shakes hands once with everyone else.

Approved solution

(a) Choose which 5 of the 12 positions hold the 1s: $\binom{12}{5} = 792$.
(b) Every path uses 5 rights and 3 ups in some order — 8 moves total. Choose which 3 are ups: $\binom{8}{3} = 56$.
(c) A handshake is an unordered pair of people: $\binom{8}{2} = 28$.

None of these looks like a selection at first. The skill is asking *what is being chosen*: positions, moves, or pairs. Once that is named, the formula is immediate.

Section D — Writing

5. In four to six complete sentences, explain why $C(n,r) = P(n,r)/r!$, and why $\binom{n}{r} = \binom{n}{n-r}$ can be seen without computing either side.

Approved solution

Model answer. Every r-element subset can be arranged in $r!$ different orders, and each of those orders is a distinct r-permutation, so the permutations count each subset exactly $r!$ times. Dividing $P(n,r)$ by $r!$ therefore corrects that over-count and leaves the number of subsets. For the second identity, choosing which r elements to include is the same act as choosing which $n-r$ to leave out: each choice determines the other completely, so the two collections of choices are in one-to-one correspondence and must have the same size. No computation is needed because the argument pairs up the objects being counted rather than evaluating the formulas.

Full credit requires the over-count explanation and the include/exclude pairing. An answer that verifies the identity algebraically earns about half — the question asks why it can be seen *without* computing.

Section E — Challenge (optional)

6. A multiple-choice test has 10 questions, each with 4 options. How many ways are there to answer the test so that exactly 7 answers are correct?

Approved solution

Choose which 7 questions are answered correctly: $\binom{10}{7} = 120$. Each of the remaining 3 must be answered *incorrectly*, and each has 3 wrong options, so $3^3 = 27$ ways.

$$\binom{10}{7} \cdot 3^3 = 120 \cdot 27 = 3240.$$

The common error is forgetting the 3^3 — there is only one right answer per question but three ways to be wrong. Choosing *which* are wrong is a combination; choosing *how* each is wrong is a

multiplication.

Instructor notes

- **Problem 4 is the set's best item.** Bit strings, grid paths, and handshakes are all $\binom{n}{r}$ once you name what is chosen. Put all three on the board together.
- **Problems 2(c) and 3(c)** both use the complement for "at least one." Students who enumerate cases get the right answer slowly and often miss one.
- **Problem 6's** 3^3 is the standard omission.
- Reserved for Homework 13: Section 8.4 and 8.5 even-numbered exercises.